
For the Restricted use in Emergency Situation by a Registered Medical Practitioner only

Hetero COVID-Vac Combined Vector Vaccine

GENERIC NAME

COVID-Vac Combined Vector Vaccine (Hetero-COVID-Vac)

COMPOSITION

Composition per dose (0.5 ml):

Component I: Recombinant adenovirus serotype 26 particles containing the SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles.

Component II: Recombinant adenovirus serotype 5 particles containing SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles.

DESCRIPTION

COVID-Vac Combined Vector Vaccine is provided as a colorless to pale yellow solution for intramuscular injection.

Component I: Recombinant adenovirus serotype 26 particles containing the SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10$ raise to 11 particles per dose of 0.5 mL

Component II: Recombinant adenovirus serotype 5 particles containing SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10$ raise to 11 particles per dose of 0.5 mL

DOSAGE FORM

A solution for intramuscular injection. Component I - 0.5 ml / dose + component II - 0.5 ml / dose.

INDICATION

The vaccine is indicated for active immunization to prevent the novel Coronavirus infection (COVID-19) in individuals of ≥ 18 years of age.

DOSAGE AND ADMINISTRATION

The Hetero-COVID-Vac Vaccine is administered intramuscularly in two doses (0.5 mL each) with interval of 21 day (Day 1: Component I & Day 21: Component II).

After the vaccine is administered, the patient should be monitored by a healthcare professional for 30 minutes.

Special populations

Elderly population

Efficacy was similar in elderly population of more than 60 years of age as compared to adults less than 60 years of age.

Paediatric population

The safety and efficacy of Hetero-COVID-Vac in children and adolescents (aged <18 years old) have not yet been established. No data are available.

Method of administration:

The vaccine is intended for intramuscular injection only. Intravenous injection of the product is strictly prohibited. The vaccine is injected into the deltoid muscle (the upper third of the outer shoulder). If it is impossible to inject into the deltoid muscle, the product is injected into the vastus lateralis muscle.

For instructions on administration

Prior to vaccination with either Component I or Component II, take a vial of the intended component out of the freezer and keep at room temperature (15-25°C) until completely thawed with no visible frozen inclusions. The vial may be held in hands to help it thaw.

Carefully mix the contents of the vial by swirling gently in an upright position for 10 seconds. Do not shake the vial.

Remove the protective plastic overlay from the vial and treat the rubber stopper with an alcohol wipe.

With a single-use syringe, draw 0.5 mL of the drug as a dose to administer to the patient.

After being thawed, the vaccine may be stored at room temperature (15-25°C) for upto 2 hours. Unused contents after this period must be discarded

USE IN SPECIAL POPULATION

It is not anticipated that there is a biologically plausible way in which the vaccine could cause infertility in any woman or man, developmental pathology, or affect offspring, since:

- The vaccine does not use adjuvants
- The potentially toxic (in rats) excipient (polysorbate 80) used in the vaccine is used in a dose that cannot affect human fertility or the reproductive function

-
- The vaccine virus does not reproduce itself; after injection, the virus delivers the S protein gene to the cell and ceases to exist in the human body
 - The gene coding S protein in the body leads to the production of the viral S protein, and the development of an immune response to it
 - Antibodies to the S protein produced in response to immunization are similar to the antibodies produced in response to a disease caused by SARS-CoV-2, therefore, the risk associated with immunization is not higher than that for infection
 - Antibodies to adenovirus produced in response to immunization are similar to antibodies to adenoviruses produced in response to a disease caused by adenovirus with a widely spread pathogen; therefore, the risk associated immunization is not higher than that for infection.
 - In preclinical studies of reproductive toxicity, a similar vaccine developed based on adenovirus vectors types 26 and 5 of identical composition was studied. No increased risk is expected with administering the drug in active reproductive populations given adherence to the restrictions indicated in the instructions for medical use.

Pregnancy

There is limited experience with use of COVID-19 Vaccine in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryo/foetal development, parturition or post-natal development (see section 5.3). Administration of COVID-19 Vaccine in pregnancy should only be considered when the potential benefits outweigh any potential risks for the mother and foetus.

Risk Summary

All pregnancies have a risk of birth defect, loss, or other adverse outcomes. The estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2% to 4% and 15% to 20%, respectively. Available data on COVID-19 Vaccine administered to pregnant women are insufficient to inform vaccine-associated risks in pregnancy.

In a developmental toxicity study, 0.2 mL of a vaccine formulation containing the same quantity of nucleoside-modified messenger ribonucleic acid (mRNA) (100 mcg) and other ingredients included in a single human dose of COVID-19 Vaccine was administered to female rats by the intramuscular route on four occasions: 28 and 14 days prior to mating, and on gestation days 1 and 13. No vaccine-related adverse effects on female fertility, fetal development or postnatal

development were reported in the study.

Lactation

It is unknown whether COVID-19 Vaccine is excreted in human milk.

Risk Summary

Data are not available to assess the effects of COVID-19 Vaccine on the breastfed infant or on milk production/excretion.

Fertility

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity.

Pediatric Use

Safety and effectiveness have not been assessed in persons less than 18 years of age. The use COVID-19 Vaccine does not include use in individuals younger than 18 years of age.

Geriatric Use

Clinical studies of COVID-19 Vaccine included participants 65 years of age and older receiving vaccine or placebo, and their data contribute to the overall assessment of safety and efficacy. In Phase 3 clinical study, 24.8% (n=7,520) of participants were 65 years of age and older and 4.6% (n=1,399) of participants were 75 years of age and older. Vaccine efficacy in participants 65 years of age and older was 86.4% (95% CI 61.4, 95.2) compared to 95.6% (95% CI 90.6, 97.9) in participants 18 to <65 years of age. Overall, there were no notable differences in the safety profiles observed in participants 65 years of age and older and younger participants.

CONTRAINDICATIONS

Contraindications for the injection of component I

- Hypersensitivity to any constituents of the vaccine.
- Severe allergic reactions in the past;
- Pregnant and lactating mothers.
- Individuals below 18 years.
- Acute infectious and non-infectious diseases, exacerbation of chronic diseases - vaccination is carried out 2-4 weeks after recovery or remission. In case of non-serious

ARVI, acute gastrointestinal infections, vaccination is carried out after the temperature has returned to normal

Contraindications for the injection of component II

- Severe post-vaccination complications (anaphylactic shock, severe generalized allergic reactions, convulsive syndrome, temperature above 40 ° C, etc.) for the injection of component I of the vaccine

WARNINGS AND PRECAUTIONS

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Hypersensitivity and anaphylaxis

Anaphylaxis has been reported. Appropriate medical treatment and supervision should always be readily available in case of an anaphylactic reaction following administration of the vaccine.

Close observation for at least 15 minutes is recommended following vaccination. The second dose of the vaccine should not be given to those who have experienced anaphylaxis to the first dose of Hetero-COVID-Vac Combined Vector Vaccine.

Anxiety-related reactions

Anxiety-related reactions, including vasovagal reactions (syncope), hyperventilation or stress-related reactions may occur in association with vaccination as a psychogenic response to the needle injection. It is important that precautions are in place to avoid injury from fainting.

Concurrent illness

Vaccination should be postponed in individuals suffering from acute severe febrile illness or acute infection. The presence of a minor infection and/or low-grade fever should not delay vaccination.

Thrombocytopenia and coagulation disorders

As with other intramuscular injections, the vaccine should be given with caution in individuals receiving anticoagulant therapy or those with thrombocytopenia or any coagulation disorder (such as haemophilia) because bleeding or bruising may occur following an intramuscular administration in these individuals.

Immunocompromised individuals

The efficacy, safety and immunogenicity of the vaccine has not been assessed in immunocompromised individuals, including those receiving immunosuppressant therapy. The efficacy of Hetero-COVID-Vac Combined Vector Vaccine may be lower in immunosuppressed individuals.

Duration of protection

The duration of protection afforded by the vaccine is unknown.

Limitations of vaccine effectiveness

Individuals may not be fully protected until 14 days after their second dose. As with all vaccines, vaccination with Hetero-COVID-Vac Combined Vector Vaccine may not protect all vaccine recipients.

The vaccine should be used with caution in cases of chronic liver and kidney disease, endocrine disorders (apparent thyroid function abnormalities and diabetes mellitus in decompensation stage), serious diseases of the hematopoietic system, epilepsy and other CNS diseases, acute coronary syndrome, and acute cerebrovascular event, myocarditis, endocarditis, pericarditis.

Due to lack of data, vaccination may be a risk for the following groups of patients:

- With autoimmune diseases (stimulation of the immune system can lead to an exacerbation of the disease, special caution should be exercised with patients with an autoimmune disorder that tend to lead to severe and life-threatening conditions);
- With malignant neoplasms.

The decision to vaccinate should be based on the assessment of the benefit/risk ratio in each specific situation.

DRUG INTERACTIONS

There are no data to assess the concomitant administration of the COVID-19 Vaccine with other vaccines.

UNDESIRABLE EFFECTS

Adverse reactions reported in a clinical trial following administration of the COVID-19 Vaccine include pain at the injection site (92.0%), fatigue (70.0%), headache (64.7%), myalgia (61.5%), arthralgia (46.4%), chills (45.4%), nausea/vomiting (23.0%), axillary swelling/tenderness

(19.8%), fever (15.5%), swelling at the injection site (14.7%), and erythema at the injection site (10.0%).

Additional adverse reactions, some of which may be serious, may become apparent with more widespread use of the COVID-19 Vaccine.

Tabulated list of adverse reactions

Adverse reactions reported are listed according to the following frequency convention: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$), very rare ($< 1/10,000$), not known (cannot be estimated from the available data)

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Blood and lymphatic system disorders

Very common: Lymphadenopathy*

Immune system disorders

Not known: Anaphylaxis, hypersensitivity

Nervous system disorders

Very common: Headache

Rare: dizziness, syncope

Rare: Acute peripheral facial, paralysis**

Gastrointestinal disorders

Very common: Nausea/vomiting

Common: dyspepsia

Skin and subcutaneous tissue disorders

Common: Rash

Musculoskeletal and connective tissue disorders

Very common: Myalgia, arthralgia

Respiratory, chest, and mediastinal disorders:

Common: oropharyngeal pain, nasal congestion, sore throat, rhinorrhea.

General disorders and administration site conditions

Very common: Injection site pain, vaccination site tenderness, fatigue, chills, pyrexia, injection site swelling, increased vaccination site skin temperature,

Common: Injection site erythema, injection site urticaria, injection site rash, hyperthermia, edema and pruritus, asthenia, malaise, decreased appetite.

Uncommon: Injection site pruritus

Rare: Facial swelling***

Lab test and instrumentation data:

divergent deviations of immunological status indicators: increased count of T-lymphocytes, increase in the percentage of lymphocytes, decreased count of natural killer cells, increased count of CD4-lymphocytes, decreased count of CD4-lymphocytes, increased count of B-lymphocytes, decreased count of B-lymphocytes, increased count of natural killer cells, increased count of CD8 lymphocytes, increased level of immunoglobulin E (IgE) in the blood, increase in the CD4/CD8 ratio, decrease in the CD4/CD8 ratio, increased level of immunoglobulin A (IgA) in the blood, decrease in the percentage of CD8 lymphocytes. Abnormalities in the complete blood count: increase in the percentage of lymphocytes, decrease in the hematocrit, increased count of lymphocytes, increase in the erythrocyte sedimentation rate, increased leukocyte count, increased count of monocytes, increased platelet count, decreased count of neutrophils, decreased platelet count. Deviations in common urine analysis: erythrocytes in the urine.

*Lymphadenopathy was captured as axillary lymphadenopathy on the same side as the injection site.

**Throughout the safety follow-up period, acute peripheral facial paralysis (or palsy) was reported by three participants in the COVID-19 Vaccine Moderna group and one participant in the placebo group. Onset in the vaccine group participants was 22 days, 28 days, and 32 days after Dose 2.

***There were two serious adverse events of facial swelling in vaccine recipients with a history of injection of dermatological fillers. The onset of swelling was reported 1 and 2 days, respectively, after vaccination

Reporting of suspected adverse reactions

Health care professionals, patients/consumers are advised to closely monitor the possibility of the above ADRs associated with the use of the above drugs. If such reactions are encountered, please report to the Hetero either by filling of **Suspect Adverse Drug Reactions Reporting Form (form.heteroworld.com)** or by **Hetero Helpline No.180-120-8689**. Also for all India

safety cases and complaints, please write to drugsafetyindia@heterodrugs.com.

OVERDOSE

No case of overdose has been reported.

Considering that the dispensing of product is allowed only for medical institutions, and the vaccination itself is carried out only by qualified medical personnel, the risk of overdose is extremely low. However, it can be assumed that with an accidental overdose, the development of the above toxic and toxic-allergic reactions to a more severe degree is possible. There are no specific antidotes to the product.

Therapeutic measures in this case will include symptomatic therapy in accordance with the indications (antipyretic /NSAID and desensitizing agents), corticosteroids - parenterally for severe toxic-allergic syndrome). The regimen for prescribing drugs should be selected according to the recommendations for use and dosages of this product. In the event of overdose, monitoring of vital functions and possible symptomatic treatment is recommended.

CLINICAL PHARMACOLOGY

Mechanism of Action

Mechanism of action COVID-19 Vaccine contains mRNA encapsulated in lipid nanoparticles. The mRNA encodes for the full-length SARS-CoV-2 spike protein modified with 2 proline substitutions within the heptad repeat 1 domain (S-2P) to stabilise the spike protein into a prefusion conformation. After intramuscular injection, cells at the injection site and the draining lymph nodes take up the lipid nanoparticle, effectively delivering the mRNA sequence into cells for translation into viral protein. The delivered mRNA does not enter the cellular nucleus or interact with the genome, is nonreplicating, and is expressed transiently mainly by dendritic cells and subcapsular sinus macrophages. The expressed, membrane-bound spike protein of SARS-CoV-2 is then recognised by immune cells as a foreign antigen. This elicits both T-cell and B-cell responses to generate neutralising antibodies, which may contribute to protection against COVID-19.

Pharmacodynamics

Immunogenicity and Vaccine effectiveness in Animals

Vaccine effectiveness and immunogenicity were studied in various animal models like mice, hamsters and primates. Hamster studies indicated that vaccination could achieve 100% survival in immunosuppressed hamsters when they are infected with SARS-COV-2 virus. Primate

studies indicated that there was significant immunogenicity developed in vaccinated animals in terms of s-glycoprotein (spike protein) specific antibodies, virus neutralizing antibody and CD4/CD8 lymphocyte proliferation.

Pharmacokinetics

Target gene expression and content analysis for adenoviral DNA were evaluated in mice administered both components of the vaccine intramuscularly in thigh muscle. The gene expression peaked on day 2 to day 14 in mice organs. The adenoviral DNAs were found restricted to the thigh muscle (Adenovirus serotype 26 and 5) and local lymphnodes (Adenovirus serotype 5) only. No other pharmacokinetic studies were conducted with the product.

INCOMPATIBILITIES

Not applicable.

STORAGE & HANDLING

Store in a light-proof place at a temperature of -18°C or below. Store in a thawed state at room temperature (15-25°C) for no more than 2 hours. Discard any unused contents after this period. Re-freezing is not allowed.

SHELF LIFE

Component- I & Component- II – 6 Months

PACKAGING INFORMATION

COVID-Vac Combined Vector Vaccine (Hetero-COVID-Vac) is presented as multi dose (3 mL) of Component I and Component II in transparent vial (type I glass) with a rubber stopper and a flip-off plastic cap with aluminium seal. Each vial of multi dose contains 5 doses of 0.5 mL per dose in (3 mL).

MANUFACTURED BY

M/s. Hetero Biopharma Limited,

Survey No. 458 (Part), TSIIC Formulation - SEZ,

Polepally Village, Jadcherla Mandal, Mahaboobnagar District – 509 301,

Telangana State, India

For the Restricted use in Emergency Situation by a Registered Medical Practitioner only

Hetero COVID-Vac Combined Vector Vaccine

1. Generic Name

COVID-Vac Combined Vector Vaccine (Hetero-COVID-Vac)

2. Qualitative and Quantitative Composition

Composition per dose (0.5 ml):

Component I: Recombinant adenovirus serotype 26 particles containing the SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles.

Component II: Recombinant adenovirus serotype 5 particles containing SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles.

3. Dosage Form & Strength

A solution for intramuscular injection. Component I - 0.5 ml / dose + component II - 0.5 ml / dose

4. Clinical Particulars

4.1 Indications

The vaccine is indicated for active immunization to prevent the novel Coronavirus infection (COVID-19) in individuals of ≥ 18 years of age.

4.2 Posology and Method of administration

The vaccine is administered intramuscularly in two doses of 0.5 mL each with interval of 21 days (Day 1: Component I & Day 21: Component II).

After the vaccine is administered, the patient should be monitored by a healthcare professional for 30 minutes.

Special populations

Elderly population

Efficacy was similar in elderly population of more than 60 years of age as compared to adults less than 60 years of age.

Paediatric population

The safety and efficacy of Hetero-COVID-Vac in children and adolescents (aged <18 years old) have not yet been established. No data are available.

Method of administration:

The vaccine is intended for intramuscular injection only. Intravenous injection of the product is strictly prohibited. The vaccine is injected into the deltoid muscle (the upper third of the outer shoulder). If it is impossible to inject into the deltoid muscle, the product is injected into the vastus lateralis muscle.

For instructions on administration

Prior to vaccination with either Component I or Component II, take a vial of the intended component out of the freezer and keep at room temperature (15-25°C) until completely thawed with no visible frozen inclusions. The vial may be held in hands to help it thaw.

Carefully mix the contents of the vial by swirling gently in an upright position for 10 seconds. Do not shake the vial.

Remove the protective plastic overlay from the vial and treat the rubber stopper with an alcohol wipe.

With a single-use syringe, draw 0.5 mL of the drug as a dose to administer to the patient.

After being thawed, the vaccine may be stored at room temperature (15-25°C) for upto 2 hours. Unused contents after this period must be discarded.

4.3 Contraindications

Contraindications for the injection of component I

- Hypersensitivity to any constituents of the vaccine.
- Severe allergic reactions in the past.
- Pregnant and lactating mothers.
- Individuals below 18 years.
- Acute infectious and non-infectious diseases, exacerbation of chronic diseases - vaccination is carried out 2-4 weeks after recovery or remission. In case of non-serious ARVI, acute gastrointestinal infections, vaccination is carried out after the temperature has returned to normal.

Contraindications for the injection of component II

- Severe post-vaccination complications (anaphylactic shock, severe generalized allergic reactions, convulsive syndrome, temperature above 40 ° C, etc.) for the injection of component I of the vaccine

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Hypersensitivity and anaphylaxis

Anaphylaxis has been reported. Appropriate medical treatment and supervision should always be readily available in case of an anaphylactic reaction following administration of the vaccine.

Close observation for at least 15 minutes is recommended following vaccination. The second dose of the vaccine should not be given to those who have experienced anaphylaxis to the first dose of Hetero-COVID-Vac Combined Vector Vaccine.

Anxiety-related reactions

Anxiety-related reactions, including vasovagal reactions (syncope), hyperventilation or stress-related reactions may occur in association with vaccination as a psychogenic response to the needle injection. It is important that precautions are in place to avoid injury from fainting.

Concurrent illness

Vaccination should be postponed in individuals suffering from acute severe febrile illness or acute infection. The presence of a minor infection and/or low-grade fever should not delay vaccination.

Thrombocytopenia and coagulation disorders

As with other intramuscular injections, the vaccine should be given with caution in individuals receiving anticoagulant therapy or those with thrombocytopenia or any coagulation disorder (such as haemophilia) because bleeding or bruising may occur following an intramuscular administration in these individuals.

Immunocompromised individuals

The efficacy, safety and immunogenicity of the vaccine has not been assessed in immunocompromised individuals, including those receiving immunosuppressant therapy. The

efficacy of Hetero-COVID-Vac Combined Vector Vaccine may be lower in immunosuppressed individuals.

Duration of protection

The duration of protection afforded by the vaccine is unknown.

Limitations of vaccine effectiveness

Individuals may not be fully protected until 14 days after their second dose. As with all vaccines, vaccination with Hetero-COVID-Vac Combined Vector Vaccine may not protect all vaccine recipients.

The vaccine should be used with caution in cases of chronic liver and kidney disease, endocrine disorders (apparent thyroid function abnormalities and diabetes mellitus in decompensation stage), serious diseases of the hematopoietic system, epilepsy and other CNS diseases, acute coronary syndrome, and acute cerebrovascular event, myocarditis, endocarditis, pericarditis.

Due to lack of data, vaccination may be a risk for the following groups of patients:

- With autoimmune diseases (stimulation of the immune system can lead to an exacerbation of the disease, special caution should be exercised with patients with an autoimmune disorder that tend to lead to severe and life-threatening conditions);
- With malignant neoplasms.

The decision to vaccinate should be based on the assessment of the benefit/risk ratio in each specific situation.

4.5 Drug Interactions

There are no data to assess the concomitant administration of the COVID-19 Vaccine with other vaccines.

4.6 Use in special populations (such as pregnant women, lactating women, paediatric patients, geriatric patients etc.)

It is not anticipated that there is a biologically plausible way in which the vaccine could cause infertility in any woman or man, developmental pathology, or affect offspring, since:

- The vaccine does not use adjuvants

- The potentially toxic (in rats) excipient (polysorbate 80) used in the vaccine is used in a dose that cannot affect human fertility or the reproductive function
- The vaccine virus does not reproduce itself; after injection, the virus delivers the S protein gene to the cell and ceases to exist in the human body
- The gene coding S protein in the body leads to the production of the viral S protein, and the development of an immune response to it
- Antibodies to the S protein produced in response to immunization are similar to the antibodies produced in response to a disease caused by SARS-CoV-2, therefore, the risk associated with immunization is not higher than that for infection
- Antibodies to adenovirus produced in response to immunization are similar to antibodies to adenoviruses produced in response to a disease caused by adenovirus with a widely spread pathogen; therefore, the risk associated immunization is not higher than that for infection.
- In preclinical studies of reproductive toxicity, a similar vaccine developed based on adenovirus vectors types 26 and 5 of identical composition was studied. No increased risk is expected with administering the drug in active reproductive populations given adherence to the restrictions indicated in the instructions for medical use.

Pregnancy

There is limited experience with use of COVID-19 Vaccine in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryo/foetal development, parturition or post-natal development (see section 5.3). Administration of COVID-19 Vaccine in pregnancy should only be considered when the potential benefits outweigh any potential risks for the mother and foetus.

Risk Summary

All pregnancies have a risk of birth defect, loss, or other adverse outcomes. The estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2% to 4% and 15% to 20%, respectively. Available data on COVID-19 Vaccine administered to pregnant women are insufficient to inform vaccine-associated risks in pregnancy.

In a developmental toxicity study, 0.2 mL of a vaccine formulation containing the same quantity of nucleoside-modified messenger ribonucleic acid (mRNA) (100 mcg) and other

ingredients included in a single human dose of COVID-19 Vaccine was administered to female rats by the intramuscular route on four occasions: 28 and 14 days prior to mating, and on gestation days 1 and 13. No vaccine-related adverse effects on female fertility, fetal development or postnatal development were reported in the study.

Lactation

It is unknown whether COVID-19 Vaccine is excreted in human milk.

Risk Summary

Data are not available to assess the effects of COVID-19 Vaccine on the breastfed infant or on milk production/excretion.

Fertility

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity.

Pediatric Use

Safety and effectiveness have not been assessed in persons less than 18 years of age. The use COVID-19 Vaccine does not include use in individuals younger than 18 years of age.

Geriatric Use

Clinical studies of COVID-19 Vaccine included participants 65 years of age and older receiving vaccine or placebo, and their data contribute to the overall assessment of safety and efficacy. In Phase 3 clinical study, 24.8% (n=7,520) of participants were 65 years of age and older and 4.6% (n=1,399) of participants were 75 years of age and older. Vaccine efficacy in participants 65 years of age and older was 86.4% (95% CI 61.4, 95.2) compared to 95.6% (95% CI 90.6, 97.9) in participants 18 to <65 years of age. Overall, there were no notable differences in the safety profiles observed in participants 65 years of age and older and younger participants.

4.7 Effects on ability to drive and use machines

COVID-19 Vaccine has no or negligible influence on the ability to drive and use machines.

However, some of the effects mentioned under section 4.8 may temporarily affect the ability to drive or use machines.

4.8 Undesirable Effects

Adverse reactions reported in a clinical trial following administration of the COVID-19 Vaccine include pain at the injection site (92.0%), fatigue (70.0%), headache (64.7%), myalgia (61.5%), arthralgia (46.4%), chills (45.4%), nausea/vomiting (23.0%), axillary swelling/tenderness (19.8%), fever (15.5%), swelling at the injection site (14.7%), and erythema at the injection site (10.0%).

Additional adverse reactions, some of which may be serious, may become apparent with more widespread use of the COVID-19 Vaccine.

Tabulated list of adverse reactions

Adverse reactions reported are listed according to the following frequency convention: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$), very rare ($< 1/10,000$), not known (cannot be estimated from the available data)

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Blood and lymphatic system disorders

Very common: Lymphadenopathy*

Immune system disorders

Not known: Anaphylaxis, hypersensitivity

Nervous system disorders

Very common: Headache

Rare: dizziness, syncope

Rare: Acute peripheral facial, paralysis**

Gastrointestinal disorders

Very common: Nausea/vomiting

Common: dyspepsia

Skin and subcutaneous tissue disorders

Common: Rash

Musculoskeletal and connective tissue disorders

Very common: Myalgia, arthralgia

Respiratory, chest, and mediastinal disorders:

Common: oropharyngeal pain, nasal congestion, sore throat, rhinorrhea.

General disorders and administration site conditions

Very common: Injection site pain, vaccination site tenderness, fatigue, chills, pyrexia, injection site swelling, increased vaccination site skin temperature,

Common: Injection site erythema, injection site urticaria, injection site rash, hyperthermia, edema and pruritus, asthenia, malaise, decreased appetite.

Uncommon: Injection site pruritus

Rare: Facial swelling***

Lab test and instrumentation data:

Divergent deviations of immunological status indicators: increased count of T-lymphocytes, increase in the percentage of lymphocytes, decreased count of natural killer cells, increased count of CD4-lymphocytes, decreased count of CD4-lymphocytes, increased count of B-lymphocytes, decreased count of B-lymphocytes, increased count of natural killer cells, increased count of CD8 lymphocytes, increased level of immunoglobulin E (IgE) in the blood, increase in the CD4/CD8 ratio, decrease in the CD4/CD8 ratio, increased level of immunoglobulin A (IgA) in the blood, decrease in the percentage of CD8 lymphocytes. Abnormalities in the complete blood count: increase in the percentage of lymphocytes, decrease in the hematocrit, increased count of lymphocytes, increase in the erythrocyte sedimentation rate, increased leukocyte count, increased count of monocytes, increased platelet count, decreased count of neutrophils, decreased platelet count. Deviations in common urine analysis: erythrocytes in the urine.

*Lymphadenopathy was captured as axillary lymphadenopathy on the same side as the injection site.

**Throughout the safety follow-up period, acute peripheral facial paralysis (or palsy) was reported by three participants in the COVID-19 Vaccine Moderna group and one participant in the placebo group. Onset in the vaccine group participants was 22 days, 28 days, and 32 days after Dose 2.

***There were two serious adverse events of facial swelling in vaccine recipients with a history of injection of dermatological fillers. The onset of swelling was reported 1 and 2 days, respectively, after vaccination

Reporting of suspected adverse reactions

Health care professionals, patients/consumers are advised to closely monitor the possibility of the above ADRs associated with the use of the above drugs. If such reactions are encountered, please report to the Hetero either by filling of **Suspect Adverse Drug Reactions Reporting Form (form.heteroworld.com)** or by **Hetero Helpline No.180-120-8689**. Also for all India safety cases and complaints, please write to drugsafetyindia@heterodrugs.com.

4.9 Overdose

No case of overdose has been reported.

Considering that the dispensing of product is allowed only for medical institutions, and the vaccination itself is carried out only by qualified medical personnel, the risk of overdose is extremely low. However, it can be assumed that with an accidental overdose, the development of the above toxic and toxic-allergic reactions to a more severe degree is possible. There are no specific antidotes to the product.

Therapeutic measures in this case will include symptomatic therapy in accordance with the indications (antipyretic /NSAID and desensitizing agents), corticosteroids - parenterally for severe toxic-allergic syndrome). The regimen for prescribing drugs should be selected according to the recommendations for use and dosages of this product. In the event of overdose, monitoring of vital functions and possible symptomatic treatment is recommended.

5. Pharmacological Properties

Pharmacotherapeutic group: medical immunobiological vaccine.

ATC code: J07B

5.1 Pharmacodynamic properties

Mechanism of action

Mechanism of action COVID-19 Vaccine contains mRNA encapsulated in lipid nanoparticles. The mRNA encodes for the full-length SARS-CoV-2 spike protein modified with 2 proline substitutions within the heptad repeat 1 domain (S-2P) to stabilise the spike protein into a prefusion conformation. After intramuscular injection, cells at the injection site and the draining lymph nodes take up the lipid nanoparticle, effectively delivering the mRNA sequence into cells for translation into viral protein. The delivered mRNA does not enter the cellular nucleus or interact with the genome, is nonreplicating, and is expressed transiently mainly by

dendritic cells and subcapsular sinus macrophages. The expressed, membrane-bound spike protein of SARS-CoV-2 is then recognised by immune cells as a foreign antigen. This elicits both T-cell and B-cell responses to generate neutralising antibodies, which may contribute to protection against COVID-19.

Immunogenicity and Vaccine effectiveness in Animals

Vaccine effectiveness and immunogenicity were studied in various animal models like mice, hamsters and primates. Hamster studies indicated that vaccination could achieve 100% survival in immunosuppressed hamsters when they are infected with SARS-COV-2 virus. Primate studies indicated that there was significant immunogenicity developed in vaccinated animals in terms of s-glycoprotein (spike protein) specific antibodies, virus neutralizing antibody and CD4/CD8 lymphocyte proliferation.

5.2 Pharmacokinetic properties

Target gene expression and content analysis for adenoviral DNA were evaluated in mice administered both components of the vaccine intramuscularly in thigh muscle. The gene expression peaked on day 2 to day 14 in mice organs. The adenoviral DNAs were found restricted to the thigh muscle (Adenovirus serotype 26 and 5) and local lymphnodes (Adenovirus serotype 5) only. No other pharmacokinetic studies were conducted with the product.

6. Nonclinical Properties

6.1 Animal Toxicology or Pharmacology

Non-clinical data reveal no special hazard for humans based on conventional studies of repeat dose toxicity and reproductive and developmental toxicity.

General Toxicity

General toxicity studies were conducted in rats (intramuscularly receiving up to 4 doses exceeding the human dose once every 2 weeks). Transient and reversible injection site oedema and erythema and transient and reversible changes in laboratory tests (including increases in eosinophils, activated partial thromboplastin time, and fibrinogen) were observed. Results suggests the toxicity potential to humans is low.

Systemic toxicity, allergenicity and immunotoxicity

Single-dose general toxicity studies were done on mice (each component separately), rabbits (components 1 and 2 in succession, with a reduced administration interval relative to planned clinical use), primates (components 1 and 2 in succession in a therapeutic dose for humans, with the administration interval that is planned for clinical use). Allergenicity tests were carried out on guinea pigs, and immunotoxicity tests on mice. There were toxicity, allergenicity or immunotoxicity was observed in this study with doses several folds high to the human equivalent dose. Studies conducted in primates also observed that there was no antibody dependent enhance reported in the vaccinated animals when they were exposed to SARS-COV-2 virus.

Genotoxicity/Carcinogenicity

In vitro and in vivo genotoxicity studies were conducted with the novel lipid component SM-102 of the vaccine. Results suggests the genotoxicity potential to humans is very low.

Carcinogenicity studies were not performed.

Reproductive Toxicity

In a developmental toxicity study, 0.2 mL of a vaccine formulation containing the same quantity of mRNA (100 micrograms) and other ingredients included in a single human dose of COVID-19 Vaccine was administered to female rats by the intramuscular route on four occasions: 28 and 14 days prior to mating, and on gestation days 1 and 13. SARS-CoV-2 antibody responses were present in maternal animals from prior to mating to the end of the study on lactation day 21 as well as in foetuses and offspring. There were no vaccine-related adverse effects on female fertility, pregnancy, embryo foetal or offspring development or postnatal development. No data are available of mRNA1273 vaccine placental transfer or excretion in milk.

COVID-Vac Combined Vector Vaccine should not be administered, if patient had experienced:

- had a severe allergic reaction after a previous dose of this vaccine
- had a severe allergic reaction to any ingredient of this vaccine

6.2 Clinical Studies

Published Literature of Sputnik V

Phase I/II Clinical Trial in Russia (NCT04436471)

38 volunteers were recruited in this trial, of which 9 each received either component 1 or 2 and were observed for 28 days thereafter as part of Phase I study. Another 20 volunteers received

component 1 followed by 2 at interval of 21 days and were followed up till day 42 (3 weeks after the second dose) as part of Phase II study. Phase I study indicated that both components of the vaccines were highly immunogenic and safe in the volunteers. Phase II indicated that humoral immunogenicity parameters s-glycoprotein (spike protein) specific antibodies and virus neutralizing antibodies increased over the observations at days, 14, 21, 28 and 42 with significantly superior titres to the convalescent plasma for the earlier parameter on days 28 and 42 as well as 100 seroconversion for both parameters by day 42. Cellular immunogenicity parameters of CD4/CD8 lymphocyte proliferation and interferon gamma secretion also increased over days 14 and 28 with 100% volunteers showing response in these parameters on day 28.

Phase III Clinical Trial in Russia (RESIST, NCT04530396)

As per the recent interim analysis, 33,771 volunteers [25,321 received Gam-COVID-Vac combined vector vaccine and 8,450 received placebo in 1:3 proportion] were enrolled in the study, which included subjects aged 18 to 92 years old (43.9 ± 12 years), 33.6% females, 8.9% subjects >60 years of age and 22.8% subjects with comorbidities.

Earlier interim analysis, which was published in Lancet. indicated that amongst 18,695 volunteers receiving both vaccine doses, there were 78 cases of COVID-19 reported [16 in vaccine arm and 62 in Placebo arm] from day 21 onwards with efficacy calculated at 91.6% (95% Confidence interval 85.6% - 95.2%). There were no significant differences in the efficacy across age groups or genders. In terms of moderate to severe COVID-19 cases, all 20 cases were reported in placebo arm indicating 100% protection against such disease. While considering the 60 COVID-19 cases reported from day 28 onwards (1 week after the second dose), the efficacy was calculated at 91.1% (95% confidence interval 83.8% to 95.1%).

In terms of humoral immunogenicity, s-glycoprotein (spike protein) specific antibody data from 980 volunteers (733 from the vaccine arm and 247 from the placebo arm), indicates that by day 42 (3 weeks after the second shot), 98.64% of volunteers in vaccine arm achieved seroconversion (with a geometric mean titre of 9818 fold) as compared to 12.55% ($P < 0.001$) volunteers in placebo arm. As per earlier interim analysis, based on 100 volunteers (72 from the vaccine arm and 28 from the placebo arm), indicates that by day 42 (3 weeks after the second shot), 95.83% of volunteers in vaccine arm achieved seroconversion (with a geometric mean titre of 44.5 fold) as compared to 7.14% ($P < 0.001$) volunteers in placebo arm. Further, on day 28 (1 week after the second dose) vaccinated arm reported significant proliferation of CD4 lymphocytes compared to CD8 lymphocytes and significant increase in interferon gamma

secretion compared to placebo arm.

Hetero Phase III Bridging Study (India; CTRI/2021/05/033665)

Phase III bridging study in India enrolled 227 subjects (170 in Hetero-COVID-Vac vaccine arm and 57 in Gam-COVID-Vac (Sputnik V) vaccine arm), of which 223 subjects were evaluated for immunogenicity parameters. SARS CoV-2 glycoprotein specific antibody data from 223 volunteers (167 in Hetero-COVID-Vac vaccine arm and 56 in Gam-COVID-Vac (Sputnik V) vaccine arm), indicates that by day 42 (3 weeks after the second shot), 99.40% of volunteers in Hetero-COVID-Vac vaccine arm achieved seroconversion (with a geometric mean titre of 12163.69 fold). For Viral neutralizing Antibody (VNA) - based on 223 volunteers (167 in Hetero-COVID-Vac vaccine arm and 56 in Gam-COVID-Vac (Sputnik V) vaccine arm), indicates that by day 28 (1 week after the second shot), 85.88% of volunteers in Hetero-COVID-Vac vaccine arm achieved seroconversion. Further, on day 28 (1 week after the second dose) Hetero-COVID-Vac vaccinated arm reported significant proliferation of CD4 lymphocytes compared to CD8 lymphocytes and significant increase in interferon gamma secretion. These results indicate the Hetero-COVID-Vac combined vector vaccine is highly immunogenic in Indian subjects.

7. Description

COVID-Vac Combined Vector Vaccine is provided as a colorless to pale yellow solution for intramuscular injection.

Component I: Recombinant adenovirus serotype 26 particles containing the SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles per dose of 0.5 mL

Component II: Recombinant adenovirus serotype 5 particles containing SARS-CoV-2 protein S gene, in the amount of $(1.0 \pm 0.5) \times 10^{11}$ particles per dose of 0.5 mL

8. Pharmaceutical Particulars

8.1 Incompatibilities

The product should not be mixed with any other medicinal products or active immunizing agents.

8.2 Shelf-life

Component- I & Component- II – 6 Months

8.3 Packing Information

COVID-Vac Combined Vector Vaccine (Hetero-COVID-Vac) is presented as multi dose (3 mL) of Component I and Component II in transparent vial (type I glass) with a rubber stopper and a flip-off plastic cap with aluminium seal. Each vial of multi dose contains 5 doses (3 mL).

8.4 Storage and handling instructions

Store in a light-proof place at a temperature of -18°C or below. Store in a thawed state at room temperature (15-25°C) for no more than 2 hours. Discard any unused contents after this period. Re-freezing is not allowed.

9. Patient Counselling Information

Side effects that have been reported with the COVID-19 Vaccine include:

- Injection site reactions: pain, tenderness and swelling of the lymph nodes in the same arm of the injection, swelling (hardness), and redness
- General side effects: fatigue, headache, muscle pain, joint pain, chills, nausea and vomiting, and fever

There is a remote chance that the COVID-19 Vaccine could cause a severe allergic reaction. A severe allergic reaction would usually occur within a few minutes to 1-hour after getting a dose of the COVID-19 Vaccine. For this reason, your vaccination provider may ask you to stay at the place where you received your vaccine for monitoring after vaccination. Signs of a severe allergic reaction can include:

- Difficulty breathing
- Swelling of your face and throat
- A fast heartbeat
- A bad rash all over your body
- Dizziness and weakness

Note: These may not be all the possible side effects of the COVID-19 Vaccine. Serious and unexpected side effects may occur. The vaccine is approved for the emergency use only.

10. Details of Manufacturer

M/s. Hetero Biopharma Limited, Survey No. 458 (Part), TSIIC Formulation -
SEZ, Polepally Village, Jadcherla Mandal, Mahaboobnagar District – 509 301, Telangana
State, India

11. Details of permission or licence number with date

MFBIO21/2000110

12. Date of revision

20-11-2021